

Source folder: /Users/jonat/Documents/GitHub/master_thesis
Files in this part: 17

File: results/dp_4_alpha_0.5_dim_64_clients_100_seed_456.json

```
-----
{
  "experiment_id": "dp_4_alpha_0.5_dim_64_clients_100_seed_456",
  "timestamp": "2026-02-11T02:34:47.725248",
  "config": {
    "num_users": 943,
    "num_items": 1682,
    "embedding_dim": 64,
    "num_clients": 100,
    "alpha": 0.5,
    "dp_epsilon": 4,
    "use_dp": true,
    "dp_sigma": 20.40234375,
    "dp_clip_norm": 1.0,
    "num_rounds": 10,
    "local_epochs": 3,
    "learning_rate": 0.01,
    "batch_size": 32,
    "seed": 456
  },
  "rounds": [
    {
      "round": 1,
      "train_loss": 227.25744999731683,
      "test_metrics": {
        "mse": 13.783183213806153,
        "mae": 3.5202975929260254,
        "samples": 20000,
        "NDCG@10": 0.058843959939489814,
        "Hit@10": 0.11,
        "Precision@10": 0.011000000000000001,
        "Recall@10": 0.0043079012461828695,
        "num_users_evaluated": 100
      },
      "aggregation": {
        "num_clients": 100,
        "total_samples": 80000
      }
    },
    {
      "round": 2,
      "train_loss": 234.6616195220771,
      "test_metrics": {
        "mse": 14.0265330078125,
        "mae": 3.5235395816802977,
        "samples": 20000
      },
      "aggregation": {
        "num_clients": 100,
        "total_samples": 80000
      }
    },
    {
      "round": 3,
      "train_loss": 240.2358970889224,
      "test_metrics": {
        "mse": 14.386404266357422,
        "mae": 3.529303728866577,
        "samples": 20000
      },
      "aggregation": {
        "num_clients": 100,
        "total_samples": 80000
      }
    },
    {
      "round": 4,
      "train_loss": 247.30434022342914,
      "test_metrics": {
        "mse": 14.87741611480713,
```

```
"mae": 3.538506480407715,
"samples": 20000
},
"aggregation": {
  "num_clients": 100,
  "total_samples": 80000
}
},
{
  "round": 5,
  "train_loss": 254.62335761036957,
  "test_metrics": {
    "mse": 15.513196069335937,
    "mae": 3.5569311714172365,
    "samples": 20000,
    "NDCG@10": 0.04920733657218897,
    "Hit@10": 0.05,
    "Precision@10": 0.005,
    "Recall@10": 0.0025243765772796157,
    "num_users_evaluated": 100
  },
  "aggregation": {
    "num_clients": 100,
    "total_samples": 80000
  }
},
{
  "round": 6,
  "train_loss": 261.4700665301408,
  "test_metrics": {
    "mse": 16.287284356689455,
    "mae": 3.5860441093444826,
    "samples": 20000
  },
  "aggregation": {
    "num_clients": 100,
    "total_samples": 80000
  }
},
{
  "round": 7,
  "train_loss": 271.48093422458817,
  "test_metrics": {
    "mse": 17.255915408325194,
    "mae": 3.636420043182373,
    "samples": 20000
  },
  "aggregation": {
    "num_clients": 100,
    "total_samples": 80000
  }
},
{
  "round": 8,
  "train_loss": 276.1572551277892,
  "test_metrics": {
    "mse": 18.261986018371584,
    "mae": 3.688884970855713,
    "samples": 20000
  },
  "aggregation": {
    "num_clients": 100,
    "total_samples": 80000
  }
},
{
  "round": 9,
  "train_loss": 282.6003147611433,
  "test_metrics": {
    "mse": 19.30410701751709,
    "mae": 3.746032738494873,
    "samples": 20000
  },
  "aggregation": {
```

```

        "num_clients": 100,
        "total_samples": 80000
    },
    {
        "round": 10,
        "train_loss": 290.2898528725529,
        "test_metrics": {
            "mse": 20.64410247039795,
            "mae": 3.8355876289367674,
            "samples": 20000,
            "NDCG@10": 0.03669679703804605,
            "Hit@10": 0.05,
            "Precision@10": 0.005,
            "Recall@10": 0.0022616144018583043,
            "num_users_evaluated": 100
        },
        "aggregation": {
            "num_clients": 100,
            "total_samples": 80000
        }
    }
],
"final_metrics": {
    "mse": 20.64410247039795,
    "mae": 3.8355876289367674,
    "samples": 20000,
    "NDCG@10": 0.03669679703804605,
    "Hit@10": 0.05,
    "Precision@10": 0.005,
    "Recall@10": 0.0022616144018583043,
    "num_users_evaluated": 100
}
}

```

File: results/dp_4_alpha_0.5_dim_64_clients_100_seed_456_summary.csv

```

Round,Train_Loss,NDCG@10,Hit@10,Precision@10,Recall@10,MSE,MAE
1,227.25744999731683,0.058843959939489814,0.11,0.011000000000000001,0.0043079012461828695,13.78318321380615
3,3.5202975929260254
2,234.6616195220771,,,,,14.0265330078125,3.5235395816802977
3,240.2358970889224,,,,,14.386404266357422,3.529303728866577
4,247.30434022342914,,,,,14.87741611480713,3.538506480407715
5,254.62335761036957,0.04920733657218897,0.05,0.005,0.0025243765772796157,15.513196069335937,3.556931171417
2365
6,261.4700665301408,,,,,16.287284356689455,3.5860441093444826
7,271.48093422458817,,,,,17.255915408325194,3.636420043182373
8,276.1572551277892,,,,,18.261986018371584,3.688884970855713
9,282.6003147611433,,,,,19.30410701751709,3.746032738494873
10,290.2898528725529,0.03669679703804605,0.05,0.005,0.0022616144018583043,20.64410247039795,3.8355876289367
674
FINAL,,0.03669679703804605,0.05,0.005,0.0022616144018583043,20.64410247039795,3.8355876289367674

```

File: results/dp_8_alpha_0.5_dim_64_clients_100_seed_123.json

```

{
  "experiment_id": "dp_8_alpha_0.5_dim_64_clients_100_seed_123",
  "timestamp": "2026-02-11T02:17:59.276389",
  "config": {
    "num_users": 943,
    "num_items": 1682,
    "embedding_dim": 64,
    "num_clients": 100,
    "alpha": 0.5,
    "dp_epsilon": 8,
    "use_dp": true,
    "dp_sigma": 11.03203125,
    "dp_clip_norm": 1.0,
    "num_rounds": 10,
    "local_epochs": 3,
    "learning_rate": 0.01,
    "batch_size": 32,
    "seed": 123
  },

```

```
"rounds": [
  {
    "round": 1,
    "train_loss": 32.2485865604616,
    "test_metrics": {
      "mse": 13.706314096069336,
      "mae": 3.5218424324035644,
      "samples": 20000,
      "NDCG@10": 0.056472994498314326,
      "Hit@10": 0.06,
      "Precision@10": 0.006000000000000001,
      "Recall@10": 0.0016320618662437314,
      "num_users_evaluated": 100
    },
    "aggregation": {
      "num_clients": 100,
      "total_samples": 80000
    }
  },
  {
    "round": 2,
    "train_loss": 33.034346703759404,
    "test_metrics": {
      "mse": 13.729706134033203,
      "mae": 3.5210181625366213,
      "samples": 20000
    },
    "aggregation": {
      "num_clients": 100,
      "total_samples": 80000
    }
  },
  {
    "round": 3,
    "train_loss": 33.56999220028218,
    "test_metrics": {
      "mse": 13.764635670471192,
      "mae": 3.520541397857666,
      "samples": 20000
    },
    "aggregation": {
      "num_clients": 100,
      "total_samples": 80000
    }
  },
  {
    "round": 4,
    "train_loss": 34.37596866098962,
    "test_metrics": {
      "mse": 13.807001486206055,
      "mae": 3.519665375137329,
      "samples": 20000
    },
    "aggregation": {
      "num_clients": 100,
      "total_samples": 80000
    }
  },
  {
    "round": 5,
    "train_loss": 35.117718829367234,
    "test_metrics": {
      "mse": 13.88942387084961,
      "mae": 3.521537496948242,
      "samples": 20000,
      "NDCG@10": 0.04564447310757874,
      "Hit@10": 0.08,
      "Precision@10": 0.008,
      "Recall@10": 0.005146976538473407,
      "num_users_evaluated": 100
    },
    "aggregation": {
      "num_clients": 100,
      "total_samples": 80000
    }
  }
]
```

```
}
},
{
  "round": 6,
  "train_loss": 35.47923740492363,
  "test_metrics": {
    "mse": 13.97281569519043,
    "mae": 3.5236828666687012,
    "samples": 20000
  },
  "aggregation": {
    "num_clients": 100,
    "total_samples": 80000
  }
},
{
  "round": 7,
  "train_loss": 36.279913027801626,
  "test_metrics": {
    "mse": 14.046265171813966,
    "mae": 3.5218750690460205,
    "samples": 20000
  },
  "aggregation": {
    "num_clients": 100,
    "total_samples": 80000
  }
},
{
  "round": 8,
  "train_loss": 36.62263538561832,
  "test_metrics": {
    "mse": 14.13695319519043,
    "mae": 3.5216548866271973,
    "samples": 20000
  },
  "aggregation": {
    "num_clients": 100,
    "total_samples": 80000
  }
},
{
  "round": 9,
  "train_loss": 37.43999281625264,
  "test_metrics": {
    "mse": 14.212836094665528,
    "mae": 3.5201156791687014,
    "samples": 20000
  },
  "aggregation": {
    "num_clients": 100,
    "total_samples": 80000
  }
},
{
  "round": 10,
  "train_loss": 37.8632549337846,
  "test_metrics": {
    "mse": 14.355028085327149,
    "mae": 3.524582919692993,
    "samples": 20000,
    "NDCG@10": 0.05788043055404635,
    "Hit@10": 0.07,
    "Precision@10": 0.006999999999999999,
    "Recall@10": 0.004431035991506316,
    "num_users_evaluated": 100
  },
  "aggregation": {
    "num_clients": 100,
    "total_samples": 80000
  }
}
],
"final_metrics": {
```

```

    "mse": 14.355028085327149,
    "mae": 3.524582919692993,
    "samples": 20000,
    "NDCG@10": 0.05788043055404635,
    "Hit@10": 0.07,
    "Precision@10": 0.006999999999999999,
    "Recall@10": 0.004431035991506316,
    "num_users_evaluated": 100
  }
}

```

File: results/dp_8_alpha_0.5_dim_64_clients_100_seed_123_summary.csv

```

Round,Train_Loss,NDCG@10,Hit@10,Precision@10,Recall@10,MSE,MAE
1,32.2485865604616,0.056472994498314326,0.06,0.006000000000000001,0.0016320618662437314,13.706314096069336,
3.5218424324035644
2,33.034346703759404,,,,,13.729706134033203,3.5210181625366213
3,33.56999220028218,,,,,13.764635670471192,3.520541397857666
4,34.37596866098962,,,,,13.807001486206055,3.519665375137329
5,35.117718829367234,0.04564447310757874,0.08,0.008,0.005146976538473407,13.88942387084961,3.52153749694824
2
6,35.47923740492363,,,,,13.97281569519043,3.5236828666687012
7,36.279913027801626,,,,,14.046265171813966,3.5218750690460205
8,36.62263538561832,,,,,14.13695319519043,3.5216548866271973
9,37.43999281625264,,,,,14.212836094665528,3.5201156791687014
10,37.8632549337846,0.05788043055404635,0.07,0.006999999999999999,0.004431035991506316,14.355028085327149,3
.524582919692993
FINAL,,0.05788043055404635,0.07,0.006999999999999999,0.004431035991506316,14.355028085327149,3.524582919692
993

```

File: results/dp_8_alpha_0.5_dim_64_clients_100_seed_42.json

```

{
  "experiment_id": "dp_8_alpha_0.5_dim_64_clients_100_seed_42",
  "timestamp": "2026-02-11T02:14:30.202965",
  "config": {
    "num_users": 943,
    "num_items": 1682,
    "embedding_dim": 64,
    "num_clients": 100,
    "alpha": 0.5,
    "dp_epsilon": 8,
    "use_dp": true,
    "dp_sigma": 11.03203125,
    "dp_clip_norm": 1.0,
    "num_rounds": 10,
    "local_epochs": 3,
    "learning_rate": 0.01,
    "batch_size": 32,
    "seed": 42
  },
  "rounds": [
    {
      "round": 1,
      "train_loss": 32.46484567453981,
      "test_metrics": {
        "mse": 13.70007797088623,
        "mae": 3.521318423843384,
        "samples": 20000,
        "NDCG@10": 0.07062999616437177,
        "Hit@10": 0.1,
        "Precision@10": 0.01,
        "Recall@10": 0.004255373351798951,
        "num_users_evaluated": 100
      },
      "aggregation": {
        "num_clients": 100,
        "total_samples": 80000
      }
    },
    {
      "round": 2,
      "train_loss": 32.984528741289346,
      "test_metrics": {

```

```
"mse": 13.713029345703125,
"mae": 3.5190051342010498,
"samples": 20000
},
"aggregation": {
  "num_clients": 100,
  "total_samples": 80000
}
},
{
  "round": 3,
  "train_loss": 33.73158305571247,
  "test_metrics": {
    "mse": 13.760207579040527,
    "mae": 3.520247960281372,
    "samples": 20000
  },
  "aggregation": {
    "num_clients": 100,
    "total_samples": 80000
  }
},
{
  "round": 4,
  "train_loss": 34.126739832006564,
  "test_metrics": {
    "mse": 13.82516392211914,
    "mae": 3.522406580734253,
    "samples": 20000
  },
  "aggregation": {
    "num_clients": 100,
    "total_samples": 80000
  }
},
{
  "round": 5,
  "train_loss": 34.842418864136576,
  "test_metrics": {
    "mse": 13.877694001770019,
    "mae": 3.5213635154724123,
    "samples": 20000,
    "NDCG@10": 0.038458220330608306,
    "Hit@10": 0.07,
    "Precision@10": 0.0070000000000000001,
    "Recall@10": 0.0038556439145193927,
    "num_users_evaluated": 100
  },
  "aggregation": {
    "num_clients": 100,
    "total_samples": 80000
  }
},
{
  "round": 6,
  "train_loss": 35.2034369765747,
  "test_metrics": {
    "mse": 13.941607377624512,
    "mae": 3.520322141265869,
    "samples": 20000
  },
  "aggregation": {
    "num_clients": 100,
    "total_samples": 80000
  }
},
{
  "round": 7,
  "train_loss": 36.004213745426235,
  "test_metrics": {
    "mse": 14.01173528137207,
    "mae": 3.5188413883209226,
    "samples": 20000
  },
  "aggregation": {
    "num_clients": 100,
    "total_samples": 80000
  }
}
```

```
"aggregation": {
  "num_clients": 100,
  "total_samples": 80000
},
{
  "round": 8,
  "train_loss": 36.656923031942206,
  "test_metrics": {
    "mse": 14.06484306640625,
    "mae": 3.5142020988464355,
    "samples": 20000
  },
  "aggregation": {
    "num_clients": 100,
    "total_samples": 80000
  }
},
{
  "round": 9,
  "train_loss": 37.13889551900223,
  "test_metrics": {
    "mse": 14.185741499328614,
    "mae": 3.517760985183716,
    "samples": 20000
  },
  "aggregation": {
    "num_clients": 100,
    "total_samples": 80000
  }
},
{
  "round": 10,
  "train_loss": 38.064433839840845,
  "test_metrics": {
    "mse": 14.309214770507813,
    "mae": 3.5198187145233155,
    "samples": 20000,
    "NDCG@10": 0.06680998371750531,
    "Hit@10": 0.06,
    "Precision@10": 0.006,
    "Recall@10": 0.0017234382593935339,
    "num_users_evaluated": 100
  },
  "aggregation": {
    "num_clients": 100,
    "total_samples": 80000
  }
}
],
"final_metrics": {
  "mse": 14.309214770507813,
  "mae": 3.5198187145233155,
  "samples": 20000,
  "NDCG@10": 0.06680998371750531,
  "Hit@10": 0.06,
  "Precision@10": 0.006,
  "Recall@10": 0.0017234382593935339,
  "num_users_evaluated": 100
},
"attack_results": {
  "mia": {
    "train_accuracy": 1.0,
    "auc": 0.53355,
    "accuracy": 0.515,
    "true_positive_rate": 0.6,
    "false_positive_rate": 0.57
  },
  "inversion": {
    "top_k_accuracy": 0.0,
    "top_k": 10,
    "num_users_evaluated": 50
  }
}
```



```
}
```

File: results/dp_8_alpha_0.5_dim_64_clients_100_seed_42_summary.csv

```
Round,Train_Loss,NDCG@10,Hit@10,Precision@10,Recall@10,MSE,MAE
1,32.46484567453981,0.07062999616437177,0.1,0.01,0.004255373351798951,13.70007797088623,3.521318423843384
2,32.984528741289346,,,,,13.713029345703125,3.5190051342010498
3,33.73158305571247,,,,,13.760207579040527,3.520247960281372
4,34.126739832006564,,,,,13.82516392211914,3.522406580734253
5,34.842418864136576,0.038458220330608306,0.07,0.007000000000000001,0.0038556439145193927,13.87769400177001
9,3.5213635154724123
6,35.2034369765747,,,,,13.941607377624512,3.520322141265869
7,36.004213745426235,,,,,14.01173528137207,3.5188413883209226
8,36.656923031942206,,,,,14.06484306640625,3.5142020988464355
9,37.13889551900223,,,,,14.185741499328614,3.517760985183716
10,38.064433839840845,0.06680998371750531,0.06,0.006,0.0017234382593935339,14.309214770507813,3.51981871452
33155
FINAL,,0.06680998371750531,0.06,0.006,0.0017234382593935339,14.309214770507813,3.5198187145233155
```

File: results/dp_8_alpha_0.5_dim_64_clients_100_seed_456.json

```
{
  "experiment_id": "dp_8_alpha_0.5_dim_64_clients_100_seed_456",
  "timestamp": "2026-02-11T02:22:26.614168",
  "config": {
    "num_users": 943,
    "num_items": 1682,
    "embedding_dim": 64,
    "num_clients": 100,
    "alpha": 0.5,
    "dp_epsilon": 8,
    "use_dp": true,
    "dp_sigma": 11.03203125,
    "dp_clip_norm": 1.0,
    "num_rounds": 10,
    "local_epochs": 3,
    "learning_rate": 0.01,
    "batch_size": 32,
    "seed": 456
  },
  "rounds": [
    {
      "round": 1,
      "train_loss": 32.29933751890424,
      "test_metrics": {
        "mse": 13.70424580230713,
        "mae": 3.5219111457824708,
        "samples": 20000,
        "NDCG@10": 0.04900664415170795,
        "Hit@10": 0.07,
        "Precision@10": 0.007000000000000001,
        "Recall@10": 0.0020227477003257922,
        "num_users_evaluated": 100
      },
      "aggregation": {
        "num_clients": 100,
        "total_samples": 80000
      }
    },
    {
      "round": 2,
      "train_loss": 32.88257406573976,
      "test_metrics": {
        "mse": 13.735208779907227,
        "mae": 3.522513412475586,
        "samples": 20000
      },
      "aggregation": {
        "num_clients": 100,
        "total_samples": 80000
      }
    },
    {
      "round": 3,
```

```
"train_loss": 33.490608923782005,
"test_metrics": {
  "mse": 13.773474458312988,
  "mae": 3.52244154586792,
  "samples": 20000
},
"aggregation": {
  "num_clients": 100,
  "total_samples": 80000
}
},
{
  "round": 4,
  "train_loss": 34.03537829007678,
  "test_metrics": {
    "mse": 13.8231083984375,
    "mae": 3.5225142669677734,
    "samples": 20000
  },
  "aggregation": {
    "num_clients": 100,
    "total_samples": 80000
  }
},
{
  "round": 5,
  "train_loss": 34.5790928287218,
  "test_metrics": {
    "mse": 13.886728021240234,
    "mae": 3.522496696472168,
    "samples": 20000,
    "NDCG@10": 0.06690983654996545,
    "Hit@10": 0.09,
    "Precision@10": 0.011000000000000001,
    "Recall@10": 0.007239654355057395,
    "num_users_evaluated": 100
  },
  "aggregation": {
    "num_clients": 100,
    "total_samples": 80000
  }
},
{
  "round": 6,
  "train_loss": 35.31948951689446,
  "test_metrics": {
    "mse": 13.964869668579102,
    "mae": 3.5228670364379884,
    "samples": 20000
  },
  "aggregation": {
    "num_clients": 100,
    "total_samples": 80000
  }
},
{
  "round": 7,
  "train_loss": 36.126836900261964,
  "test_metrics": {
    "mse": 14.081374354553223,
    "mae": 3.5267831939697265,
    "samples": 20000
  },
  "aggregation": {
    "num_clients": 100,
    "total_samples": 80000
  }
},
{
  "round": 8,
  "train_loss": 36.630888192507676,
  "test_metrics": {
    "mse": 14.17117515258789,
    "mae": 3.5257721771240234,
```

```

    "samples": 20000
  },
  "aggregation": {
    "num_clients": 100,
    "total_samples": 80000
  }
},
{
  "round": 9,
  "train_loss": 36.98664217515131,
  "test_metrics": {
    "mse": 14.257159707641602,
    "mae": 3.5239249969482422,
    "samples": 20000
  },
  "aggregation": {
    "num_clients": 100,
    "total_samples": 80000
  }
},
{
  "round": 10,
  "train_loss": 37.696265293993996,
  "test_metrics": {
    "mse": 14.396677098083495,
    "mae": 3.529047723388672,
    "samples": 20000,
    "NDCG@10": 0.03561013735332017,
    "Hit@10": 0.05,
    "Precision@10": 0.005,
    "Recall@10": 0.001212587264121489,
    "num_users_evaluated": 100
  },
  "aggregation": {
    "num_clients": 100,
    "total_samples": 80000
  }
}
],
"final_metrics": {
  "mse": 14.396677098083495,
  "mae": 3.529047723388672,
  "samples": 20000,
  "NDCG@10": 0.03561013735332017,
  "Hit@10": 0.05,
  "Precision@10": 0.005,
  "Recall@10": 0.001212587264121489,
  "num_users_evaluated": 100
}
}

```

File: results/dp_8_alpha_0.5_dim_64_clients_100_seed_456_summary.csv

```

Round,Train_Loss,NDCG@10,Hit@10,Precision@10,Recall@10,MSE,MAE
1,32.29933751890424,0.04900664415170795,0.07,0.007000000000000001,0.0020227477003257922,13.70424580230713,3.5219111457824708
2,32.88257406573976,,,,,13.735208779907227,3.522513412475586
3,33.490608923782005,,,,,13.773474458312988,3.52244154586792
4,34.03537829007678,,,,,13.8231083984375,3.5225142669677734
5,34.5790928287218,0.06690983654996545,0.09,0.011000000000000001,0.007239654355057395,13.886728021240234,3.522496696472168
6,35.31948951689446,,,,,13.964869668579102,3.5228670364379884
7,36.126836900261964,,,,,14.081374354553223,3.5267831939697265
8,36.630888192507676,,,,,14.17117515258789,3.5257721771240234
9,36.98664217515131,,,,,14.257159707641602,3.5239249969482422
10,37.696265293993996,0.03561013735332017,0.05,0.005,0.001212587264121489,14.396677098083495,3.529047723388672
FINAL,,0.03561013735332017,0.05,0.005,0.001212587264121489,14.396677098083495,3.529047723388672

```

File: results/dp_inf_alpha_0.1_dim_64_clients_100_seed_123.json

```

{
  "experiment_id": "dp_inf_alpha_0.1_dim_64_clients_100_seed_123",
  "timestamp": "2026-02-11T02:58:54.374272",

```

```
"config": {
  "num_users": 943,
  "num_items": 1682,
  "embedding_dim": 64,
  "num_clients": 100,
  "alpha": 0.1,
  "dp_epsilon": null,
  "use_dp": false,
  "dp_sigma": 0.0,
  "dp_clip_norm": 1.0,
  "num_rounds": 10,
  "local_epochs": 3,
  "learning_rate": 0.01,
  "batch_size": 32,
  "seed": 123
},
"rounds": [
  {
    "round": 1,
    "train_loss": 13.736760905458635,
    "test_metrics": {
      "mse": 13.68513099822998,
      "mae": 3.5215539653778074,
      "samples": 20000,
      "NDCG@10": 0.039105890351920955,
      "Hit@10": 0.04,
      "Precision@10": 0.004,
      "Recall@10": 0.0023614718614718613,
      "num_users_evaluated": 100
    },
    "aggregation": {
      "num_clients": 100,
      "total_samples": 80000
    }
  },
  {
    "round": 2,
    "train_loss": 13.73491219975885,
    "test_metrics": {
      "mse": 13.685111654663086,
      "mae": 3.5215516830444336,
      "samples": 20000
    },
    "aggregation": {
      "num_clients": 100,
      "total_samples": 80000
    }
  },
  {
    "round": 3,
    "train_loss": 13.74112209662515,
    "test_metrics": {
      "mse": 13.685089517211914,
      "mae": 3.521549138259888,
      "samples": 20000
    },
    "aggregation": {
      "num_clients": 100,
      "total_samples": 80000
    }
  },
  {
    "round": 4,
    "train_loss": 13.740059274786885,
    "test_metrics": {
      "mse": 13.685065866088868,
      "mae": 3.521546283340454,
      "samples": 20000
    },
    "aggregation": {
      "num_clients": 100,
      "total_samples": 80000
    }
  }
],
```

```
{
  "round": 5,
  "train_loss": 13.731428745961571,
  "test_metrics": {
    "mse": 13.685040350341797,
    "mae": 3.5215431350708006,
    "samples": 20000,
    "NDCG@10": 0.039105890351920955,
    "Hit@10": 0.04,
    "Precision@10": 0.004,
    "Recall@10": 0.0023614718614718613,
    "num_users_evaluated": 100
  },
  "aggregation": {
    "num_clients": 100,
    "total_samples": 80000
  }
},
{
  "round": 6,
  "train_loss": 13.726476204674901,
  "test_metrics": {
    "mse": 13.68501517791748,
    "mae": 3.5215401008605958,
    "samples": 20000
  },
  "aggregation": {
    "num_clients": 100,
    "total_samples": 80000
  }
},
{
  "round": 7,
  "train_loss": 13.731320599997662,
  "test_metrics": {
    "mse": 13.684988584899902,
    "mae": 3.521536766433716,
    "samples": 20000
  },
  "aggregation": {
    "num_clients": 100,
    "total_samples": 80000
  }
},
{
  "round": 8,
  "train_loss": 13.72987025365621,
  "test_metrics": {
    "mse": 13.684961358642578,
    "mae": 3.5215334129333495,
    "samples": 20000
  },
  "aggregation": {
    "num_clients": 100,
    "total_samples": 80000
  }
},
{
  "round": 9,
  "train_loss": 13.726996088726253,
  "test_metrics": {
    "mse": 13.684932322692871,
    "mae": 3.5215298175811767,
    "samples": 20000
  },
  "aggregation": {
    "num_clients": 100,
    "total_samples": 80000
  }
},
{
  "round": 10,
  "train_loss": 13.727156604630522,
  "test_metrics": {
```

```

        "mse": 13.68490050048828,
        "mae": 3.5215259605407714,
        "samples": 20000,
        "NDCG@10": 0.039105890351920955,
        "Hit@10": 0.04,
        "Precision@10": 0.004,
        "Recall@10": 0.0023614718614718613,
        "num_users_evaluated": 100
    },
    "aggregation": {
        "num_clients": 100,
        "total_samples": 80000
    }
}
],
"final_metrics": {
    "mse": 13.68490050048828,
    "mae": 3.5215259605407714,
    "samples": 20000,
    "NDCG@10": 0.039105890351920955,
    "Hit@10": 0.04,
    "Precision@10": 0.004,
    "Recall@10": 0.0023614718614718613,
    "num_users_evaluated": 100
}
}

```

File: results/dp_inf_alpha_0.1_dim_64_clients_100_seed_123_summary.csv

```

Round,Train_Loss,NDCG@10,Hit@10,Precision@10,Recall@10,MSE,MAE
1,13.736760905458635,0.039105890351920955,0.04,0.004,0.0023614718614718613,13.68513099822998,3.521553965377
8074
2,13.73491219975885,,,,,13.685111654663086,3.5215516830444336
3,13.74112209662515,,,,,13.685089517211914,3.521549138259888
4,13.740059274786885,,,,,13.685065866088868,3.521546283340454
5,13.731428745961571,0.039105890351920955,0.04,0.004,0.0023614718614718613,13.685040350341797,3.52154313507
08006
6,13.726476204674901,,,,,13.68501517791748,3.5215401008605958
7,13.731320599997662,,,,,13.684988584899902,3.521536766433716
8,13.72987025365621,,,,,13.684961358642578,3.5215334129333495
9,13.726996088726253,,,,,13.684932322692871,3.5215298175811767
10,13.727156604630522,0.039105890351920955,0.04,0.004,0.0023614718614718613,13.68490050048828,3.52152596054
07714
FINAL,,0.039105890351920955,0.04,0.004,0.0023614718614718613,13.68490050048828,3.5215259605407714

```

File: results/dp_inf_alpha_0.1_dim_64_clients_100_seed_42.json

```

{
  "experiment_id": "dp_inf_alpha_0.1_dim_64_clients_100_seed_42",
  "timestamp": "2026-02-11T02:57:15.878026",
  "config": {
    "num_users": 943,
    "num_items": 1682,
    "embedding_dim": 64,
    "num_clients": 100,
    "alpha": 0.1,
    "dp_epsilon": null,
    "use_dp": false,
    "dp_sigma": 0.0,
    "dp_clip_norm": 1.0,
    "num_rounds": 10,
    "local_epochs": 3,
    "learning_rate": 0.01,
    "batch_size": 32,
    "seed": 42
  },
  "rounds": [
    {
      "round": 1,
      "train_loss": 13.733752253548612,
      "test_metrics": {
        "mse": 13.685993676757812,
        "mae": 3.5218222717285155,
        "samples": 20000,

```

```
"NDCG@10": 0.062217199771718815,
"Hit@10": 0.06,
"Precision@10": 0.006000000000000001,
"Recall@10": 0.004123646725745421,
"num_users_evaluated": 100
},
"aggregation": {
  "num_clients": 100,
  "total_samples": 80000
}
},
{
  "round": 2,
  "train_loss": 13.734872546745589,
  "test_metrics": {
    "mse": 13.686004968261718,
    "mae": 3.5218238746643067,
    "samples": 20000
  },
  "aggregation": {
    "num_clients": 100,
    "total_samples": 80000
  }
},
{
  "round": 3,
  "train_loss": 13.737935499853727,
  "test_metrics": {
    "mse": 13.686016731262207,
    "mae": 3.521825635910034,
    "samples": 20000
  },
  "aggregation": {
    "num_clients": 100,
    "total_samples": 80000
  }
},
{
  "round": 4,
  "train_loss": 13.732044025551165,
  "test_metrics": {
    "mse": 13.686027822875976,
    "mae": 3.5218271327972412,
    "samples": 20000
  },
  "aggregation": {
    "num_clients": 100,
    "total_samples": 80000
  }
},
{
  "round": 5,
  "train_loss": 13.72877357683148,
  "test_metrics": {
    "mse": 13.686038833618165,
    "mae": 3.5218287628173828,
    "samples": 20000,
    "NDCG@10": 0.06462270726815646,
    "Hit@10": 0.06,
    "Precision@10": 0.006000000000000001,
    "Recall@10": 0.004123646725745421,
    "num_users_evaluated": 100
  },
  "aggregation": {
    "num_clients": 100,
    "total_samples": 80000
  }
},
{
  "round": 6,
  "train_loss": 13.735028349771538,
  "test_metrics": {
    "mse": 13.686049507141114,
    "mae": 3.5218302474975585,
```

```
    "samples": 20000
  },
  "aggregation": {
    "num_clients": 100,
    "total_samples": 80000
  }
},
{
  "round": 7,
  "train_loss": 13.734060697267678,
  "test_metrics": {
    "mse": 13.686059539794922,
    "mae": 3.5218316707611086,
    "samples": 20000
  },
  "aggregation": {
    "num_clients": 100,
    "total_samples": 80000
  }
},
{
  "round": 8,
  "train_loss": 13.726528907965097,
  "test_metrics": {
    "mse": 13.686069602966308,
    "mae": 3.5218331199645996,
    "samples": 20000
  },
  "aggregation": {
    "num_clients": 100,
    "total_samples": 80000
  }
},
{
  "round": 9,
  "train_loss": 13.732529271134387,
  "test_metrics": {
    "mse": 13.686078648376466,
    "mae": 3.521834460067749,
    "samples": 20000
  },
  "aggregation": {
    "num_clients": 100,
    "total_samples": 80000
  }
},
{
  "round": 10,
  "train_loss": 13.727845996262907,
  "test_metrics": {
    "mse": 13.686086491394043,
    "mae": 3.5218355697631836,
    "samples": 20000,
    "NDCG@10": 0.06462270726815646,
    "Hit@10": 0.06,
    "Precision@10": 0.006000000000000001,
    "Recall@10": 0.004123646725745421,
    "num_users_evaluated": 100
  },
  "aggregation": {
    "num_clients": 100,
    "total_samples": 80000
  }
}
],
"final_metrics": {
  "mse": 13.686086491394043,
  "mae": 3.5218355697631836,
  "samples": 20000,
  "NDCG@10": 0.06462270726815646,
  "Hit@10": 0.06,
  "Precision@10": 0.006000000000000001,
  "Recall@10": 0.004123646725745421,
  "num_users_evaluated": 100
}
```



```
}
}
```

File: results/dp_inf_alpha_0.1_dim_64_clients_100_seed_42_summary.csv

```
Round,Train_Loss,NDCG@10,Hit@10,Precision@10,Recall@10,MSE,MAE
1,13.733752253548612,0.062217199771718815,0.06,0.006000000000000001,0.004123646725745421,13.685993676757812
,3.5218222717285155
2,13.734872546745589,,,,,13.686004968261718,3.5218238746643067
3,13.737935499853727,,,,,13.686016731262207,3.521825635910034
4,13.732044025551165,,,,,13.686027822875976,3.5218271327972412
5,13.72877357683148,0.06462270726815646,0.06,0.006000000000000001,0.004123646725745421,13.686038833618165,3
.5218287628173828
6,13.735028349771538,,,,,13.686049507141114,3.5218302474975585
7,13.734060697267678,,,,,13.686059539794922,3.5218316707611086
8,13.726528907965097,,,,,13.686069602966308,3.5218331199645996
9,13.732529271134387,,,,,13.686078648376466,3.521834460067749
10,13.727845996262907,0.06462270726815646,0.06,0.006000000000000001,0.004123646725745421,13.686086491394043
,3.5218355697631836
FINAL,,0.06462270726815646,0.06,0.006000000000000001,0.004123646725745421,13.686086491394043,3.521835569763
1836
```

File: results/dp_inf_alpha_0.1_dim_64_clients_100_seed_456.json

```
{
  "experiment_id": "dp_inf_alpha_0.1_dim_64_clients_100_seed_456",
  "timestamp": "2026-02-11T03:00:33.121918",
  "config": {
    "num_users": 943,
    "num_items": 1682,
    "embedding_dim": 64,
    "num_clients": 100,
    "alpha": 0.1,
    "dp_epsilon": null,
    "use_dp": false,
    "dp_sigma": 0.0,
    "dp_clip_norm": 1.0,
    "num_rounds": 10,
    "local_epochs": 3,
    "learning_rate": 0.01,
    "batch_size": 32,
    "seed": 456
  },
  "rounds": [
    {
      "round": 1,
      "train_loss": 13.739092155123583,
      "test_metrics": {
        "mse": 13.688507695007324,
        "mae": 3.522353038024902,
        "samples": 20000,
        "NDCG@10": 0.0577968700397596,
        "Hit@10": 0.09,
        "Precision@10": 0.009,
        "Recall@10": 0.0037602713721248985,
        "num_users_evaluated": 100
      },
      "aggregation": {
        "num_clients": 100,
        "total_samples": 80000
      }
    },
    {
      "round": 2,
      "train_loss": 13.732709845652328,
      "test_metrics": {
        "mse": 13.688491870117188,
        "mae": 3.5223504791259765,
        "samples": 20000
      },
      "aggregation": {
        "num_clients": 100,
        "total_samples": 80000
      }
    }
  ]
}
```

```
},
{
  "round": 3,
  "train_loss": 13.730281041256085,
  "test_metrics": {
    "mse": 13.688476957702637,
    "mae": 3.522347919845581,
    "samples": 20000
  },
  "aggregation": {
    "num_clients": 100,
    "total_samples": 80000
  }
},
{
  "round": 4,
  "train_loss": 13.733265482019936,
  "test_metrics": {
    "mse": 13.688458750915528,
    "mae": 3.5223448657989502,
    "samples": 20000
  },
  "aggregation": {
    "num_clients": 100,
    "total_samples": 80000
  }
},
{
  "round": 5,
  "train_loss": 13.736973903328403,
  "test_metrics": {
    "mse": 13.688440277099609,
    "mae": 3.5223418727874756,
    "samples": 20000,
    "NDCG@10": 0.0577968700397596,
    "Hit@10": 0.09,
    "Precision@10": 0.009,
    "Recall@10": 0.0037602713721248985,
    "num_users_evaluated": 100
  },
  "aggregation": {
    "num_clients": 100,
    "total_samples": 80000
  }
},
{
  "round": 6,
  "train_loss": 13.737321372608362,
  "test_metrics": {
    "mse": 13.688421134948731,
    "mae": 3.5223387226104736,
    "samples": 20000
  },
  "aggregation": {
    "num_clients": 100,
    "total_samples": 80000
  }
},
{
  "round": 7,
  "train_loss": 13.730427924763708,
  "test_metrics": {
    "mse": 13.688401861572265,
    "mae": 3.5223355705261232,
    "samples": 20000
  },
  "aggregation": {
    "num_clients": 100,
    "total_samples": 80000
  }
},
{
  "round": 8,
  "train_loss": 13.730882347794372,
```

```

    "test_metrics": {
      "mse": 13.68837836151123,
      "mae": 3.522331908798218,
      "samples": 20000
    },
    "aggregation": {
      "num_clients": 100,
      "total_samples": 80000
    }
  },
  {
    "round": 9,
    "train_loss": 13.728986479967666,
    "test_metrics": {
      "mse": 13.688355522155762,
      "mae": 3.5223282726287843,
      "samples": 20000
    },
    "aggregation": {
      "num_clients": 100,
      "total_samples": 80000
    }
  },
  {
    "round": 10,
    "train_loss": 13.73102463094966,
    "test_metrics": {
      "mse": 13.688331869506836,
      "mae": 3.522324473571777,
      "samples": 20000,
      "NDCG@10": 0.0577968700397596,
      "Hit@10": 0.09,
      "Precision@10": 0.009,
      "Recall@10": 0.0037602713721248985,
      "num_users_evaluated": 100
    },
    "aggregation": {
      "num_clients": 100,
      "total_samples": 80000
    }
  }
],
"final_metrics": {
  "mse": 13.688331869506836,
  "mae": 3.522324473571777,
  "samples": 20000,
  "NDCG@10": 0.0577968700397596,
  "Hit@10": 0.09,
  "Precision@10": 0.009,
  "Recall@10": 0.0037602713721248985,
  "num_users_evaluated": 100
}
}

```

File: results/dp_inf_alpha_0.1_dim_64_clients_100_seed_456_summary.csv

```

Round,Train_Loss,NDCG@10,Hit@10,Precision@10,Recall@10,MSE,MAE
1,13.739092155123583,0.0577968700397596,0.09,0.009,0.0037602713721248985,13.688507695007324,3.5223530380249
02
2,13.732709845652328,,,,,13.688491870117188,3.5223504791259765
3,13.730281041256085,,,,,13.688476957702637,3.522347919845581
4,13.733265482019936,,,,,13.688458750915528,3.5223448657989502
5,13.736973903328403,0.0577968700397596,0.09,0.009,0.0037602713721248985,13.688440277099609,3.5223418727874
756
6,13.737321372608362,,,,,13.688421134948731,3.5223387226104736
7,13.730427924763708,,,,,13.688401861572265,3.5223355705261232
8,13.730882347794372,,,,,13.68837836151123,3.522331908798218
9,13.728986479967666,,,,,13.688355522155762,3.5223282726287843
10,13.73102463094966,0.0577968700397596,0.09,0.009,0.0037602713721248985,13.688331869506836,3.5223244735717
77
FINAL,,0.0577968700397596,0.09,0.009,0.0037602713721248985,13.688331869506836,3.522324473571777

```

File: results/dp_inf_alpha_0.5_dim_64_clients_100_seed_123.json

```
{
  "experiment_id": "dp_inf_alpha_0.5_dim_64_clients_100_seed_123",
  "timestamp": "2026-02-11T02:09:30.261904",
  "config": {
    "num_users": 943,
    "num_items": 1682,
    "embedding_dim": 64,
    "num_clients": 100,
    "alpha": 0.5,
    "dp_epsilon": null,
    "use_dp": false,
    "dp_sigma": 0.0,
    "dp_clip_norm": 1.0,
    "num_rounds": 10,
    "local_epochs": 3,
    "learning_rate": 0.01,
    "batch_size": 32,
    "seed": 123
  },
  "rounds": [
    {
      "round": 1,
      "train_loss": 13.72604914243546,
      "test_metrics": {
        "mse": 13.685133767700195,
        "mae": 3.521554398727417,
        "samples": 20000,
        "NDCG@10": 0.039105890351920955,
        "Hit@10": 0.04,
        "Precision@10": 0.004,
        "Recall@10": 0.0023614718614718613,
        "num_users_evaluated": 100
      },
      "aggregation": {
        "num_clients": 100,
        "total_samples": 80000
      }
    },
    {
      "round": 2,
      "train_loss": 13.721697556470144,
      "test_metrics": {
        "mse": 13.685116149902344,
        "mae": 3.5215523292541504,
        "samples": 20000
      },
      "aggregation": {
        "num_clients": 100,
        "total_samples": 80000
      }
    },
    {
      "round": 3,
      "train_loss": 13.716468174163426,
      "test_metrics": {
        "mse": 13.685094821166992,
        "mae": 3.521549824142456,
        "samples": 20000
      },
      "aggregation": {
        "num_clients": 100,
        "total_samples": 80000
      }
    },
    {
      "round": 4,
      "train_loss": 13.72688887396247,
      "test_metrics": {
        "mse": 13.68507374572754,
        "mae": 3.521547329711914,
        "samples": 20000
      }
    }
  ],
}
```

```
"aggregation": {
  "num_clients": 100,
  "total_samples": 80000
},
{
  "round": 5,
  "train_loss": 13.7228971265283,
  "test_metrics": {
    "mse": 13.685051594543458,
    "mae": 3.5215446388244627,
    "samples": 20000,
    "NDCG@10": 0.039105890351920955,
    "Hit@10": 0.04,
    "Precision@10": 0.004,
    "Recall@10": 0.0023614718614718613,
    "num_users_evaluated": 100
  },
  "aggregation": {
    "num_clients": 100,
    "total_samples": 80000
  }
},
{
  "round": 6,
  "train_loss": 13.731610071622235,
  "test_metrics": {
    "mse": 13.685030517578125,
    "mae": 3.521542232513428,
    "samples": 20000
  },
  "aggregation": {
    "num_clients": 100,
    "total_samples": 80000
  }
},
{
  "round": 7,
  "train_loss": 13.714259418429437,
  "test_metrics": {
    "mse": 13.685006246948243,
    "mae": 3.521539258956909,
    "samples": 20000
  },
  "aggregation": {
    "num_clients": 100,
    "total_samples": 80000
  }
},
{
  "round": 8,
  "train_loss": 13.722299991741497,
  "test_metrics": {
    "mse": 13.68498218383789,
    "mae": 3.5215363761901854,
    "samples": 20000
  },
  "aggregation": {
    "num_clients": 100,
    "total_samples": 80000
  }
},
{
  "round": 9,
  "train_loss": 13.716793745279674,
  "test_metrics": {
    "mse": 13.684957339477538,
    "mae": 3.521533363723755,
    "samples": 20000
  },
  "aggregation": {
    "num_clients": 100,
    "total_samples": 80000
  }
}
```

```

    },
    {
      "round": 10,
      "train_loss": 13.716583412823606,
      "test_metrics": {
        "mse": 13.684931239318848,
        "mae": 3.5215301780700683,
        "samples": 20000,
        "NDCG@10": 0.039105890351920955,
        "Hit@10": 0.04,
        "Precision@10": 0.004,
        "Recall@10": 0.0023614718614718613,
        "num_users_evaluated": 100
      },
      "aggregation": {
        "num_clients": 100,
        "total_samples": 80000
      }
    }
  ],
  "final_metrics": {
    "mse": 13.684931239318848,
    "mae": 3.5215301780700683,
    "samples": 20000,
    "NDCG@10": 0.039105890351920955,
    "Hit@10": 0.04,
    "Precision@10": 0.004,
    "Recall@10": 0.0023614718614718613,
    "num_users_evaluated": 100
  }
}

```

File: results/dp_inf_alpha_0.5_dim_64_clients_100_seed_123_summary.csv

```

Round,Train_Loss,NDCG@10,Hit@10,Precision@10,Recall@10,MSE,MAE
1,13.72604914243546,0.039105890351920955,0.04,0.004,0.0023614718614718613,13.685133767700195,3.521554398727
417
2,13.721697556470144,,,,,13.685116149902344,3.5215523292541504
3,13.716468174163426,,,,,13.685094821166992,3.521549824142456
4,13.72688887396247,,,,,13.68507374572754,3.521547329711914
5,13.7228971265283,0.039105890351920955,0.04,0.004,0.0023614718614718613,13.685051594543458,3.5215446388244
627
6,13.731610071622235,,,,,13.685030517578125,3.521542232513428
7,13.714259418429437,,,,,13.685006246948243,3.521539258956909
8,13.722299991741497,,,,,13.68498218383789,3.5215363761901854
9,13.716793745279674,,,,,13.684957339477538,3.521533363723755
10,13.716583412823606,0.039105890351920955,0.04,0.004,0.0023614718614718613,13.684931239318848,3.5215301780
700683
FINAL,,0.039105890351920955,0.04,0.004,0.0023614718614718613,13.684931239318848,3.5215301780700683

```

File: results/dp_inf_alpha_0.5_dim_64_clients_100_seed_42.json

```

{
  "experiment_id": "dp_inf_alpha_0.5_dim_64_clients_100_seed_42",
  "timestamp": "2026-02-11T02:07:40.304687",
  "config": {
    "num_users": 943,
    "num_items": 1682,
    "embedding_dim": 64,
    "num_clients": 100,
    "alpha": 0.5,
    "dp_epsilon": null,
    "use_dp": false,
    "dp_sigma": 0.0,
    "dp_clip_norm": 1.0,
    "num_rounds": 10,
    "local_epochs": 3,
    "learning_rate": 0.01,
    "batch_size": 32,
    "seed": 42
  },
  "rounds": [
    {
      "round": 1,

```

```

"train_loss": 13.716136363102116,
"test_metrics": {
  "mse": 13.685996994018554,
  "mae": 3.5218226345062256,
  "samples": 20000,
  "NDCG@10": 0.06191074357045362,
  "Hit@10": 0.06,
  "Precision@10": 0.0060000000000000001,
  "Recall@10": 0.004123646725745421,
  "num_users_evaluated": 100
},
"aggregation": {
  "num_clients": 100,
  "total_samples": 80000
}
},
{
  "round": 2,
  "train_loss": 13.72660259513691,
  "test_metrics": {
    "mse": 13.686008929443359,
    "mae": 3.521824362945557,
    "samples": 20000
  },
  "aggregation": {
    "num_clients": 100,
    "total_samples": 80000
  }
},
{
  "round": 3,
  "train_loss": 13.709877476344737,
  "test_metrics": {
    "mse": 13.686023374938966,
    "mae": 3.521826485824585,
    "samples": 20000
  },
  "aggregation": {
    "num_clients": 100,
    "total_samples": 80000
  }
},
{
  "round": 4,
  "train_loss": 13.717380628722836,
  "test_metrics": {
    "mse": 13.686035110473632,
    "mae": 3.521828102493286,
    "samples": 20000
  },
  "aggregation": {
    "num_clients": 100,
    "total_samples": 80000
  }
},
{
  "round": 5,
  "train_loss": 13.722983389297813,
  "test_metrics": {
    "mse": 13.686049906921387,
    "mae": 3.521830167007446,
    "samples": 20000,
    "NDCG@10": 0.06462270726815646,
    "Hit@10": 0.06,
    "Precision@10": 0.0060000000000000001,
    "Recall@10": 0.004123646725745421,
    "num_users_evaluated": 100
  },
  "aggregation": {
    "num_clients": 100,
    "total_samples": 80000
  }
}
},
{

```

```
"round": 6,
"train_loss": 13.715259929181203,
"test_metrics": {
  "mse": 13.686062969970703,
  "mae": 3.5218320011138915,
  "samples": 20000
},
"aggregation": {
  "num_clients": 100,
  "total_samples": 80000
}
},
{
  "round": 7,
  "train_loss": 13.711035469856222,
  "test_metrics": {
    "mse": 13.686077563476562,
    "mae": 3.5218339515686035,
    "samples": 20000
  },
  "aggregation": {
    "num_clients": 100,
    "total_samples": 80000
  }
}
},
{
  "round": 8,
  "train_loss": 13.717059906595148,
  "test_metrics": {
    "mse": 13.68608907470703,
    "mae": 3.521835577774048,
    "samples": 20000
  },
  "aggregation": {
    "num_clients": 100,
    "total_samples": 80000
  }
}
},
{
  "round": 9,
  "train_loss": 13.726437523744785,
  "test_metrics": {
    "mse": 13.686100819396973,
    "mae": 3.5218372089385985,
    "samples": 20000
  },
  "aggregation": {
    "num_clients": 100,
    "total_samples": 80000
  }
}
},
{
  "round": 10,
  "train_loss": 13.716294993644347,
  "test_metrics": {
    "mse": 13.686113897705079,
    "mae": 3.5218390003204347,
    "samples": 20000,
    "NDCG@10": 0.06462270726815646,
    "Hit@10": 0.06,
    "Precision@10": 0.006000000000000001,
    "Recall@10": 0.004123646725745421,
    "num_users_evaluated": 100
  },
  "aggregation": {
    "num_clients": 100,
    "total_samples": 80000
  }
}
},
"final_metrics": {
  "mse": 13.686113897705079,
  "mae": 3.5218390003204347,
  "samples": 20000,
```



```
"NDCG@10": 0.06462270726815646,  
"Hit@10": 0.06,  
"Precision@10": 0.006000000000000001,  
"Recall@10": 0.004123646725745421,  
"num_users_evaluated": 100  
},  
"attack_results": {  
  "mia": {  
    "train_accuracy": 1.0,  
    "auc": 0.54805,  
    "accuracy": 0.545,  
    "true_positive_rate": 0.48,  
    "false_positive_rate": 0.39  
  },  
  "inversion": {  
    "top_k_accuracy": 0.0,  
    "top_k": 10,  
    "num_users_evaluated": 50  
  }  
}
```